Kone Tshivhinda

AI Systems Engineer | Software Engineer & Full-Stack Developer | Agentic Automation

Johannesburg, South Africa | erictshivhinda@gmail.com | **Portfolio:** konethegreat.github.io
GitHub (primary): github.com/konethegreatest | **GitHub (personal and coursework):** github.com/konethegreat
LinkedIn: linkedin.com/in/kone-tshivhinda-32a760233 | **Credly:** credly.com/users/kone-tshivhinda/badges
Nationality: South African | Driver's Licence: Code C1

PROFESSIONAL SUMMARY

AI systems engineer and full-stack developer who ships agent-driven software from first commit to production. Founding engineer at Motsoeneng Bill (law, forensics and advisory), leading multi-tenant, POPIA-aligned platforms built on Anthropic Claude, hybrid RAG, MCP tool use and human-in-the-loop controls. Daily user of every major AI coding tool (Claude Code, OpenAI Codex, GitHub Copilot, Gemini, Cursor, Kiro, Antigravity) and hands-on with local and sovereign LLMs (Ollama, LM Studio), embeddings, evals and LLM cost control. 20+ projects delivered across AI, legal-tech, civic-tech and enterprise software, five with live deployments. Administer the firm's GitHub organisation, AWS estate and AI keys.

AI ENGINEERING SKILLS

Agentic AI: Multi-agent orchestration, loop engineering (plan, act, verify, retry), supervisor and sub-agents, tool and function calling, Model Context Protocol (MCP) hosts and servers, human-in-the-loop sign-off, agent governance (CLAUDE.md, AGENTS.md, ADRs).

RAG & Knowledge: Hybrid dense and sparse retrieval fused by Reciprocal Rank Fusion, pgvector, embeddings (Voyage AI, nomic-embed-text), grounded and cited answers, document AI (Docling, OCR), entity resolution, Neo4j knowledge graphs.

LLM Engineering: Prompt engineering, prompt caching, Batch API, adaptive thinking, structured outputs and JSON-schema contracts, capability-tier model routing, cheap-first cascades, usage metering and cost circuit-breakers.

Evaluation & Safety: Offline LLM evals and provider-quality baselines, prompt-injection fencing, source attribution, never-fabricate guardrails, POPIA-aware data handling and data residency.

Models & APIs: Anthropic Claude (Opus, Sonnet; API and SDK), OpenAI (GPT API, Codex), Gemini, Voyage AI, IBM Watson NLP; open-weight models on Ollama and LM Studio (Llama 3.1); local LLM fine-tuning and on-prem deployment.

AI Dev Tools: Claude Code, OpenAI Codex, GitHub Copilot, Gemini, Cursor, Kiro (spec-driven development), Google Antigravity, Kimi, VS Code AI Toolkit, custom Claude Skills, n8n workflow automation, Postman.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, C#, SQL, HTML/CSS, PHP, PowerShell, Bash.

Back-End & APIs: FastAPI, Flask, Django, Spring Boot, Spring Modulith, Node.js, Express, NestJS, REST, GraphQL, WebSockets, OAuth2/OIDC, JWT, Keycloak, Prisma, SQLAlchemy.

Front-End & Mobile: React, Next.js, Vite, Angular, Vue, Tailwind CSS, Bootstrap, shadcn/ui, Redux Toolkit, HTMX, React Native (Expo), PWAs.

Data: PostgreSQL (pgvector, row-level security, schema-per-tenant), Neo4j, Redis/Valkey, MySQL, SQLite, MongoDB, Supabase, Neon, Firebase.

Cloud & DevOps: AWS (ECS Fargate, RDS, S3, CloudFront, ALB, ElastiCache, ECR, Secrets Manager, EventBridge, SES/SNS, CloudWatch, EC2), Terraform, CloudFormation, Docker, GitHub Actions (OIDC), nginx, Vercel, Render, Netlify, Azure, GCP.

Security & Quality: RBAC, MFA, rate limiting, CSP, secret scanning (gitleaks), SBOM and CVE scanning, Dependabot, pytest, Jest, Vitest, Playwright, Testcontainers, Sentry.

Tools & Practice: Git, GitHub, GitLab, VS Code, IntelliJ, PyCharm, npm, pnpm, Yarn, monorepos, ADRs, agile delivery; technical leadership, mentoring, problem-solving, collaboration.

PROFESSIONAL EXPERIENCE

Software Engineer

June 2026 - Present

Motsoeneng Bill (Technology Division) | Johannesburg, South Africa

- **Founding engineer, six-member team:** Spearhead the setup, structure and growth of a new software engineering department inside a law, forensics and advisory firm; project lead on scalable multi-tenant enterprise applications with schema-per-tenant isolation, row-level security and POPIA-aligned compliance.
- **Agentic AI and automation:** Design and run multi-agent pipelines using loop engineering (plan, act, verify, retry across delegated sub-agents) with supervisor agents, tool calling, MCP and human sign-off, eliminating manual bottlenecks across tender, due-diligence and invoicing workflows. Spec-driven development with CLAUDE.md and AGENTS.md instruction files, ADRs and audit trails keeps AI-assisted code reviewable.

- **Production AI platforms:** Primary developer on six systems (Sentinel, Tender Intelligence, MB IQ, Attest, Fee Invoice Analysis, delivery telemetry). Claude and Voyage AI sit behind owned provider seams with prompt caching, Batch API, tiered model routing, usage metering and hard cost circuit-breakers; hybrid RAG with grounded, cited answers and offline evals for prompt-injection defence and source attribution.
- **Local and sovereign AI:** Train, fine-tune and deploy local LLMs (Ollama, LM Studio) for privacy-sensitive work; researched on-premise LLM infrastructure (GPU sizing, POPIA, Johannesburg colocation, open-weight stacks); built API-key-based AI integrations for internal tooling.
- **Cloud, security and administration:** Administer the GitHub organisation, the AWS estate (ECS Fargate, RDS, S3, CloudFront, EC2 with SSM-only access) and all critical developer accounts and AI API keys (Anthropic, Voyage AI and others); Terraform, CloudFormation and GitHub Actions (OIDC) pipelines with secret scanning, SBOM and CVE gates.
- **Application management and leadership:** Oversee production uptime, security and performance; work with senior members and shareholders to deliver legal-sector software with a view to expanding across industries. Developed the keynote "The Great Shift: AGI and Africa's Next Century" for a legal-sector audience (September 2026).

Founder
2025 - Present

K-ONE IT SOLUTIONS | Johannesburg, South Africa

- Founded an independent IT and software studio; delivered the company website (HTML, CSS, PHP), a React social-media platform and a Dockerised server-health monitoring tool (Python) with Slack and email alerting.

Intern Assistant Platoon Leader
July 2024 - August 2024

GAO Tek Inc. | Canada (Remote)

- Directed operations for multiple technical squads, ensuring seamless coordination and high productivity.
- Mentored and provided technical support to interns so project benchmarks were met.
- Collaborated with senior management to deliver data-driven reports on squad performance and project milestones.

Intern Assistant Squad Leader & Tech Support
April 2024 - July 2024

GAO Tek Inc. | Canada (Remote)

- Managed quality control by reviewing technical descriptions and code and giving constructive feedback.
- Facilitated team meetings to align technical execution with management's strategic goals.
- Onboarded new interns on tools and workflow; managed WordPress product descriptions with 100% accuracy.

PROJECTS

Primary GitHub (most active): github.com/konethegreatest. Private repositories (including all Motsoeneng Bill Technology work): samples on request; live systems are linked. Personal and coursework: github.com/konethegreat. Case studies, status labels and live links: konethegreat.github.io.

AI Platforms at Motsoeneng Bill Technology

Sentinel - Forensics Due Diligence System
2026

Primary developer | Python, FastAPI, Next.js, PostgreSQL + pgvector, Neo4j, AWS | **Live:** d15nal0vd3dmw6.cloudfront.net

Agentic integrity-intelligence platform that sweeps 20+ sanctions, PEP, registry and media sources, resolves entities into a graph and delivers cited, sign-off-gated due-diligence reports.

- Deterministic entity graph plus a multi-agent state machine (NLP aggregation, entity resolution, adversarial review); RAG "AI interrogation" chat answers only from indexed sources; human sign-off, immutable audit trail, MFA/SSO and usage budgets.
- 20+ integrations (OFAC, UN, EU and UK-HMT sanctions, PEP lists, Interpol, CIPC, SAFLII, UK Companies House, SEC EDGAR); about 900 source files, 78 migrations, 148 test files and an offline AI eval baseline; Terraform-managed AWS infrastructure in af-south-1.

Tender Intelligence Platform
2026

Primary developer | Python, FastAPI, PostgreSQL + pgvector, React, Claude, Voyage AI, AWS | Private repository

Multi-tenant, agent-driven SaaS that finds, reads, qualifies and assembles public-sector tenders, from acquisition to a ready-to-authorise submission pack.

- Six-stage pipeline: multi-source acquisition (South Africa, World Bank, AfDB, UNGM), Docling PDF/OCR understanding, cross-source de-duplication, a cheap-first keyword, embedding and LLM relevance cascade, win-probability scoring and Excel exports.
- Document Vault with AI-read expiry dates, CV bank and staffing matcher, submission packs (Draft, Ready, Authorised; never auto-submits), grounded tool-calling assistant, learning loop from bid outcomes and a hard LLM cost circuit-breaker. Schema-per-tenant PostgreSQL, six-role RBAC and POPIA consent gate; shipped through release R22 (30+ modules, offline-capable Expo field app).

MB IQ - AI-Native Engineering Command Centre 2026

Solo build | Next.js 16, React 19, Prisma, PostgreSQL + pgvector, Claude, MCP, CloudFormation | Private platform | **Public v0.6 demo:** github.com/konethegreat/MB-IQ

Engineering command centre where Claude-powered agents observe, architect, review and document delivery.

- Six specialised agents (observer, architect, reviewer, spec-decomposer, presentation, supervisor) on Claude with four capability tiers and no hard-coded model IDs, adaptive thinking, prompt caching, Batch API and tool search; MCP host plus an MCP endpoint exposing 13 tools.
- Hybrid RAG (pgvector + tsvector fused by Reciprocal Rank Fusion), GitHub control plane (webhooks and reconciliation), Yjs CRDT collaborative docs, delivery analytics, multi-tenant RBAC, Sentry, CSP, AES-256-GCM secrets, seven scheduled EventBridge jobs and a seven-job CI pipeline.
- Earlier MB IQ v0.6 prototype (React 18, Express, Prisma, SQLite): five-tier RBAC dashboards, an MCP host with server-enforced human-in-the-loop approvals, Risk and Delay Radar, meeting intelligence and Diagram Studio.

Attest - Legal Careers With Evidence 2026

Solo build | NestJS, Next.js, Prisma, PostgreSQL, Claude, pnpm monorepo | Private repository, in development

Career platform for South African lawyers: verified credentials, salary-transparent jobs and explainable checklists instead of black-box match scores.

- Career Passport, Open Badges v2 credentials with public verification, reviewer-assisted LPC, LPFF and SAQA register checks, 30-day salary reconfirmation, video interviews, confidential conflict reviews, CPD tracking, admin verification queue and audit log; consent-based Claude drafting with a hard rule that AI never verifies credentials or decides hiring.

Fee Invoice Analysis Tool 2026

Primary developer | Java 21, Spring Boot, PostgreSQL, Python FastAPI, Claude, Terraform, AWS | Internal tool

Decodes glyph-obfuscated legal invoice PDFs, reconciles every fee line to the cent, auto-categorises with learned rules and Claude, and exports the team's Excel analysis.

- Python extractor (PyMuPDF, fontTools) decodes obfuscated glyphs and reconciles totals on two axes; Claude-backed categorisation replaced a regex classifier, alongside learned fingerprint rules and an export gate. Spring Boot, Thymeleaf/HTMX and Apache POI app on AWS (ECS Fargate, RDS, ALB, CloudFront) via Terraform, with CloudWatch alarms and SNS alerts.

Personal AI Projects

Vigil - Intelligence and Operations for Africa Sep 2026

Solo build (konethegreatest) | Java, Python, TypeScript, Next.js, PostgreSQL, Ollama | Private repository

Sovereign operational-intelligence platform that models evidence, investigations and missions as a governed ontology, designed so no prompt ever leaves South Africa.

- Java core, a Python "Mind" service (extraction, planner, JSON-schema contracts) with optional local Ollama reasoner and challenger models, and a Next.js cockpit; model proposals must quote exact evidence and pass analyst review before publication.
- Locally verifiable audit chain, evidence-linked plans, idempotent and concurrency-checked commands, 12 ADRs, POPIA s71 human-decision controls and an INSTRUCTIONS.md compliance contract that binds humans and AI agents; 43-phase roadmap; built with Claude and OpenAI Codex agents.

South African Job Agent Suite (SA-JAS) May - Jun 2026

Solo build | Electron, React, Express, Playwright, Claude, Gemini, OpenRouter | **Code (public snapshot):** github.com/konethegreat/scratch

Desktop app where three AI agents find real South African vacancies, tailor an ATS-optimised CV and cover letter for each role, and drive a visible browser copilot through the application, always leaving the final Submit to the user.

- Job Hunter: Claude grounded web search returns real vacancies with source URLs, backed by 12+ SA job-board scrapers and AI link validation and repair. Apply Copilot: per-page AI planner, vision fallback, three-pass smart-fill and a learn-as-you-go answer bank; never auto-submits, never solves CAPTCHAs, scans pages for prompt injection.
- Evolving candidate memory with "Dreaming" curation, unattended scheduled Routines, opt-in Anthropic computer-use form filling, parallel sub-agents for concurrent tailoring and research, a referral pipeline and cost controls (usage ledger, daily budget guard); local-first data, a threat model and 51 commits in under three weeks.

Attorney Performance Management System (PMS) May - Jun 2026

Solo build | React 18, Vite, Node.js, Express, PostgreSQL, Prisma, JWT, Claude | **Code (demo snapshot, synthetic data):** github.com/konethegreat/mostoeneng-pms

360-degree performance-review platform for a law firm: attorneys launch a review cycle, managers, peers, subordinates and clients give feedback, and approved, immutable quarterly reports roll up into an annual report.

- Optional Claude layer that is always labelled, editable and never auto-finalises: review organiser, pre-submit bias and quality check, memory-aware streamed report-draft assistant and a persistent attorney memory, with every call logged

for cost and audit. Four roles, SLA nudges, notifications and branded PDF exports; POPIA-minded design (server-side proxies, hashed avatar seeds, IDOR fix); built spec-first with CLAUDE.md guidance files; 31 commits.

Diary - AI-Augmented Personal Life Log

Jun 2026

Solo build | Python, FastAPI, HTMX, SQLite, Claude API, Ollama | **Code:** github.com/konethegreat/Diary

Private PWA that turns daily entries into an AI-coached life log, with RAG "ask my diary" over local embeddings.

- AI polish and mood scoring, morning brief, topic summaries, weekly review and a monthly AI coach with memory; runtime toggle between Claude API and local Ollama (nomic-embed-text); Google OAuth, PDF export, Tailscale.

Local AI Document Generator

Jun 2026

Solo build | Python, FastAPI, Ollama (Llama 3.1), ReportLab | Local project

Fully offline report writer: a FastAPI endpoint prompts a local Llama 3.1 model to draft Markdown, then renders a formatted PDF with ReportLab (sample output: a complete internship report).

Enterprise Platforms and Team-Delivered Systems (Motsoeneng Bill Technology)

Enterprise Case Management Intelligence Platform

2026

Project lead | Java 21, Spring Boot 4, Spring Modulith, PostgreSQL, Keycloak, Next.js 16, AWS | Private repository

Multi-tenant case-management platform for corporate and state-owned clients covering the full matter lifecycle, with 13 oversight reports and four role-based portals (internal, treasury, auditor and firm).

- Modular monolith of 20 modules across six domains (about 560 Java files, 63 Flyway migrations), row-level security, Keycloak with a custom Java MFA authenticator, Redis/Bucket4j rate limiting, ClamAV malware scanning, MinIO storage, Microsoft 365 email, Terraform modules and SBOM/CVE scanning in CI.

National Wills Week Registration Platform

Sep 2026

Team project | Node.js, Express, PostgreSQL, Docker, Render | **Live:** mb-wills-week.onrender.com

Public registration site for the firm's National Wills Week (14-18 September 2026) with POPIA consent capture, a JWT-secured admin dashboard, Brevo email and CSV export; Docker Compose and nginx on Render with seven-check CI.

MB Anchor Games

Sep 2026

Team project | Node.js, Express, PostgreSQL, Docker, AWS EC2 | **Live (staff access):** games.mb.co.za

Staff-engagement platform for the firm's 23 September 2026 convention: live leaderboard, three quizzes, "Who said this", QR scavenger hunt, awards survey with Excel and PowerPoint export, escape-room module and admin studio. AWS EC2 host with TLS, SSM-only access and nightly S3 backups; rebuilt from an earlier Next.js and Cloudflare prototype.

Knowledge Vault Enterprise v2.5

Aug 2026

Team project | JavaScript, Supabase (Postgres RLS, Edge Functions, Auth, Storage) | Private repository

Institutional knowledge repository and library-circulation system: waitlists, overdue and damaged-item workflows, 77-profile staff directory, digital resources, restricted case archive with chain of custody, reports, audit trail and backup.

Mandela Day "67 Minutes" Campaign Site

Jul 2026

Team project | JavaScript, Node.js, Express, Supabase | Private repository

Community-outreach and pro-bono campaign site for Mandela Day, backed by Supabase and an Express API.

Engineering-Cadence Telemetry and Organisation Profile

2026

Primary developer | Python, GitHub Actions, GitHub API | **Code:** github.com/Motsoeneng-Bill-Tech/.github

Daily pipeline that measures delivery consistency from tamper-proof, server-stamped evidence (pull-request, review and issue dates) instead of commit counts, then generates an executive dashboard, SVG graphs and a key-person-risk view, and updates the organisation profile README. Fail-loud by design.

Web Products and Civic Tech (Personal)

MuniSolve ZA - Municipal Service-Delivery Platform

Feb - Jun 2026

Personal project | React 19, Node.js, Express, PostgreSQL, Prisma, Claude | **Live:** munisolve-za.vercel.app | **Code:** github.com/konethegreat/munisolve-za

Citizens report municipal issues, administrators triage them on a live heat map, and "Siyanda", a Claude-powered assistant, answers citizen questions (rate-limited to 30 messages per hour).

- JWT, bcrypt and Google OAuth, Helmet and rate limiting, role-based access (citizen, admin, super-admin) with activity logs, Leaflet heat map, Open-Meteo and Nager.Date widgets; deployed on Vercel, Render and Neon.

TimeLex - Automated Legal Time Capture (v1 and v2)

Apr - May 2026

Personal project | Next.js, Prisma, PostgreSQL, Auth.js, Vercel, Netlify | **Live (v1):** timelex-automated-time-tracking.netlify.app | **Code:** github.com/konethegreat/timelex-automated-time-tracking-v2

Assessment prototype that auto-captures lawyers' billable time, drafts entries for approval and generates invoices.

- v1: automated capture engine, draft-and-approve workflow, autocomplete, targets dashboard, automated invoicing. v2: multi-tenant B2B SaaS with Microsoft Graph activity capture (simulated), validation queue, smart narrative engine, ledger sync-lock, pro-forma analytics with 15% VAT and a Ghost Practice integration target.

UI Vault - React Component Library

Personal project | React, Vite, Tailwind CSS, shadcn/ui | Local project

Reusable library of animated React components (Particles, SplashCursor, Vortex) for rapid, polished UI builds.

Server Health Monitor

Aug - Sep 2025

Personal project | Python, Docker | **Code:** github.com/konethegreat/server-health-monitor

Monitors CPU, memory and disk with threshold alerts to Slack and email, structured logging and Docker packaging.

K-ONE IT SOLUTIONS Web Builds

Feb 2025

Founder project | HTML, CSS, PHP, React | **Code:** github.com/konethegreat/k-one-it-solutions-website

Company website (HTML, CSS, PHP) and K-One Solutions Media (k-one-solutions-media), a React social-media platform.

- K-One Solutions Media source: github.com/konethegreat/k-one-solutions-media

Coursework Capstones (IBM Professional Certificate, 2026)

Repositories under github.com/konethegreat.

- **Emotion Detection AI App** (Flask, IBM Watson NLP EmotionPredict, unit tests, static analysis) and **Full-Stack Dealership Capstone** (Django, React, Node/Express, MongoDB, Docker, CI).
- **Online Course App** (Django on IBM Cloud Foundry), **Book Review API** (Express, JWT and session auth), **Paradise Nursery and e-plantShopping** (React 18, Redux Toolkit, Vite) and two **Git and GitHub** exercises.

EDUCATION

Diploma in Information Technology (NQF 6) | Damelin — February 2022 - December 2024

National Senior Certificate (Bachelor's Pass) | Noorderland High School — 2020

CERTIFICATIONS

- **IBM Full Stack Software Developer Professional Certificate** (12 courses, completed) | IBM / Coursera | Verify: coursera.org/account/accomplishments/specialization/ESSIQ8HCQBA6
- Full Stack Application Development Capstone Project | IBM / Coursera
- **IBM / Coursera courses:** Django Application Development with SQL and Databases; Developing AI Applications with Python and Flask; Python for Data Science, AI & Development; Developing Back-End Apps with Node.js and Express; Developing Front-End Apps with React; Introduction to Software Engineering; Introduction to Cloud Web Development; Introduction to Cloud Computing; Getting Started with Git and GitHub.
- **GAO Tek Inc.:** Certificate of Training in Tech Support; Certificate of Training in Digital Marketing; Certificate of Completion (Operational Excellence).

LEADERSHIP

- Assistant Squad Leader and Assistant Platoon Leader, WILL Programme (GAO Tek).
- Head Boy and Chess/Astronomy Captain, Laerskool Soutpansberg, fostering teamwork and event coordination.
- Led the Representative Council of Learners (RCL) at Noorderland High School, driving student initiatives.

LANGUAGES

English (Advanced) | Tshivenda (Native) | Northern Sotho (Intermediate)

REFERENCES

Available on request.